2024

سلايدات لاب إلكترونيات

د.ماجد الدويري

▼ من له شغف في طلب العلم، لا يتعبه بحث ولا يرهقه جهد، نحن أمة اقرأ.. أبناء القلم



اتحاد سنافر البوليتكنك

SPU

EXPERIMENT ONE

PN JUNCTION DIODE CHARACTERISTICS

1.1 Objectives:

To plot the V-I characteristics of a PN junction diode in both forward and reverse directions. Find cut in voltage (knee voltage), static and dynamic resistance in forward direction at forward current of 2mA & 8mA respectively. Find static and dynamic resistance at 10V in reverse bias condition.

1.2 Components & Equipment required:

#	Device	Range/Rating	Quantity (in No.s)
1.	Semiconductor diode trainer Board Containing: DC Power Supply Diode (Silicon) Diode (Germanium) Carbon Film Resistor	 (0-15) V 1N 4007 OA79 1 KΩ, 1/2 W	 1 1 1 1
2.	DC Voltmeter	(0-20) V	1
3.	DC Ammeter	(0-20) mA	1
4.	Connecting wires	5A	10

1.3 Theory:

A p-n junction diode conducts only in one direction. The V-I characteristics of the diode are curve between voltage across the diode and current through the diode. When external voltage is zero, circuit is open and the potential barrier does not allow the current to flow. Therefore, the circuit current is zero. When P-type (Anode is connected to +ve terminal and n- type (cathode) is connected to –ve terminal of the supply voltage, is known as forward bias. The potential barrier is reduced when diode is in the forward biased condition. At some forward voltage, the potential barrier altogether eliminated and current starts flowing through the diode and also in the circuit. The diode is said to be in ON state. The current increases with increasing forward voltage. When N-type (cathode) is connected to +ve terminal and P-type (Anode) is connected –ve terminal of the supply voltage is known as reverse bias and the potential barrier across the junction increases. Therefore, the junction resistance becomes very high and a very small current (reverse saturation current) flows in the circuit. The diode is said to be in OFF state. The reverse bias current due to minority charge carriers.

1.4 Procedure:

A. **Forward Bias**:

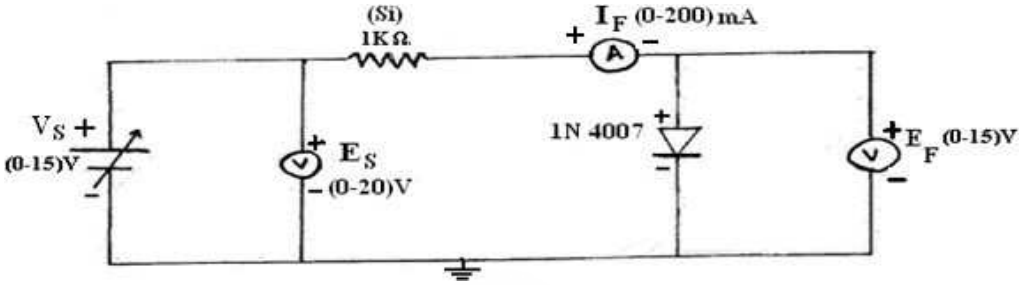
1. Connect the circuit as shown in figure (1).
2. Vary the supply voltage E_s in steps and note down the corresponding values of E_f and I_f as shown in the tabular column.

B. **Reverse Bias**:

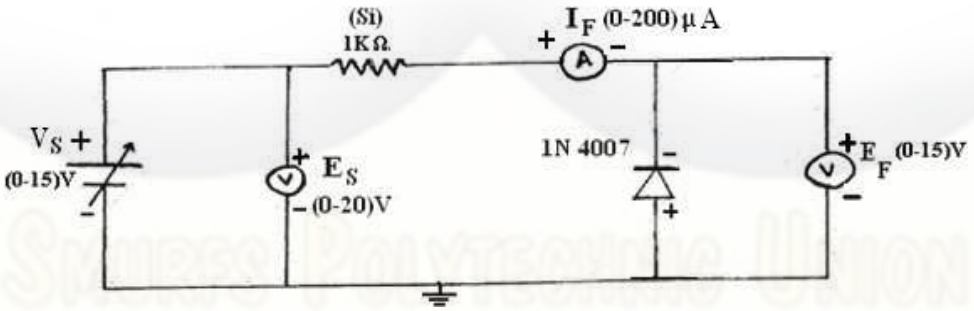
1. Connect the circuit as shown in figure (2).
2. Repeat the procedure as in forward bias and note down the corresponding Values of E_r and I_r as shown in the tabular column.

1.5 Circuit Diagrams:

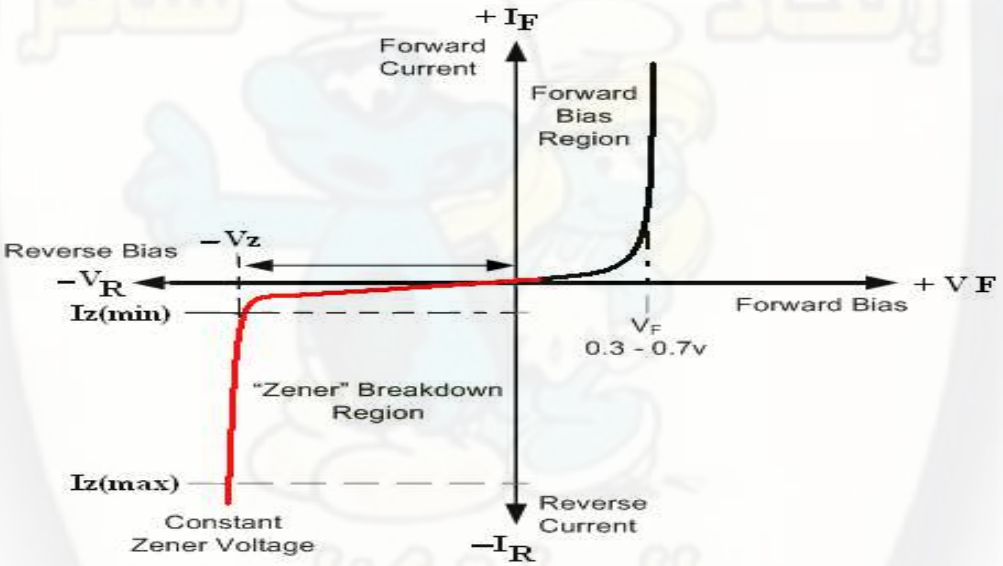
Forward Bias



Reverse Bias



1.6 Expected Graphs:



1.7 Tabular Column:

Forward Bias

E_s (volts)	E_f (volts)	I_f (mA)
0.1		
0.2		
0.3		
0.4		
0.5		
0.6		
0.7		
0.8		
0.9		
1		
2		
4		
6		
8		
10		
12		
14		

Reverse Bias

E_s (volts)	E_r (volts)	I_r (μA)
0.1		
0.2		
0.3		
0.4		
0.5		
0.6		
0.7		
0.8		
0.9		
1		
2		
4		
6		
8		
10		
12		
14		

1.8 Precautions:

1. Ensure that the polarities of the power supply and the meters as per the circuit diagram.
2. Keep the input voltage knob of the regulated power supply in minimum position both when switching ON or switching OFF the power supply.
3. No loose contacts at the junctions.
4. Ensure that the ratings of the meters are as per the circuit design for precision.

1.9 Calculations:

A. <u>Forward Bias</u>:

Static Resistance at 8mA = E_f / I_f =

Static resistance at 2mA = E_f / I_f =

Dynamic resistance at 8mA = $\Delta E_f / \Delta I_f$ =

Dynamic resistance at 8mA = $\Delta E_f / \Delta I_f$ =

B. <u>Reverse Bias</u>:

Static Resistance at (10V) = E_r / I_r =

Dynamic resistance at (10V) = $\Delta E_r / \Delta I_r$ =

1.10 Questions:

1. Define depletion region of a diode?
2. What is meant by transition & space charge capacitance of a diode?
3. Is the V-I relationship of a diode Linear or Exponential?
4. Define cut-in voltage of a diode and specify the values for Si and Ge diodes?
5. What are the applications of a p-n diode?
6. Draw the ideal characteristics of P-N junction diode?
7. What is the diode equation?
8. What is PIV?
9. What is the break down voltage?
10. What is the effect of temperature on PN junction diodes?
11. Specifications of diodes

1.11 Lab Assignment:

To plot the V-I characteristics of a PN junction (Germanium) diode in both forward and reverse directions by using multisim.

1.12 Result:

V-I characteristics of PN junction are plotted and verified in both forward and reverse directions.

Forward direction

i. Cut-in-voltage=0.7V
ii. a) Dynamic Resistance (at 8 mA) =
    b) Dynamic Resistance (at 2mA) =
iii. a) Static Resistance (at 8mA) =
iv. b) Static Resistance (at 2mA) =

Reverse Direction

i. Static Resistance (at 10V) =
ii. Dynamic Resistance (at 10 V) =

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EXPERIMENT TWO

APPLICATIONS OF DIODES

1.1 Objectives:

To verify the truth table for Logic Gates (AND & OR) using Diodes.

1.2 Components & Equipment required:

#	Device	Range/Rating	Quantity (in No.s)
01	Diode	IN4007	02
02	Resistance	1KΩ,10KΩ	01
03	Regulated power supply	(0-30V)	01
04	Voltmeter/Digital Multimeter		01
05	Breadboard and Wires		

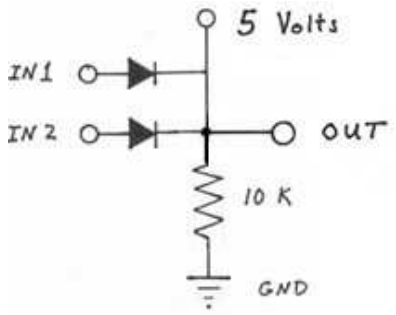
1.3 Theory:

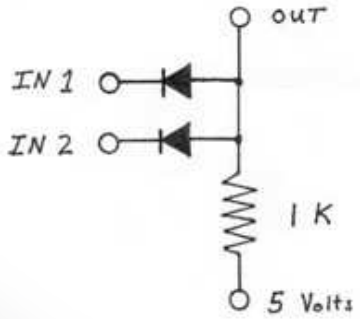
There are three types of basic Logic Operations. These can be executed by electronic circuits or devices (also called gates or switches). They are: AND, OR and NOT. AND and OR have two inputs and one output. NOT has one input and one output. Using 0's to represent 0 Volts and 1's to represent 5 Volts, the tables below show the outputs for all the possible inputs on the logic gates. Below each table (called a "truth table") is the electronic symbol for the device

1.4 Procedure:

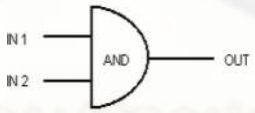
1. Connections are made as per the circuit diagram of AND gate.
2. Apply the supply of 5V
3. Give the In1 and In2 to the circuit according to the truth table (i.e for „0" apply 0V and for „1" apply 5V) and verify the output of the circuit and tabulate the readings.
4. Verify the truth table with the truth table of the AND gate.
5. Repeat the steps 1 to 4 for the circuit diagram of OR gate.

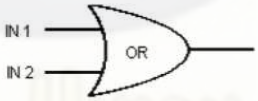
1.5 Circuit Diagrams:











DDL OR Gate **DDL AND Gate**

Truth Table:

In1	In2	Output
0	0	0
0	1	1
1	0	1
1	1	1

In1	In2	Output
0	0	0
0	1	0
1	0	0
1	1	1

1.6 Tabular Column:

Logic Gate	In1	In2	O/P
AND			

Logic Gate	In1	In2	O/P
OR			

1.7 Precautions

1. Loose and wrong connections should be avoided.
2. Supply should be switched on only after giving all the input connections.
3. Power should be switched off while connecting.

1.8 Questions:

1. Explain about the switching action of transistor
2. Draw the internal diagram of PN diode

3. Give some applications of diode.

1.9 Lab Assignment:

To verify the Truth Table.

1.10 Result:

The truth tables of the AND and OR gates using DDL logic is verified.

EXPERIMENT THREE

HALFWAVE RECTIFIERS WITH/WITHOUT FILTERS

3.1 Objectives:

Examine the input and output waveforms of a half wave rectifier without and with filters. Calculate the ripple factor with load resistance of 500Ω, 1 KΩ and 10 KΩ respectively. Calculate ripple factor with a filter capacitor of 100μF and the load of 1KΩ, 2KΩ and 10KΩ respectively.

3.2 Components & Equipment required:

#	Device	Range/Rating	Quantity (in No.s)
1	Rectifier and Filter trainer Board Containing a) AC Supply. b) Silicon Diodes c) Capacitor	 (9-0-9) V 1N 4007 0.47μF	 1 1 1
2	a) DC Voltmeter b) AC Voltmeter	(0-20) V (0-20) V	1 1
3	DC Ammeter	(0-50) mA	1
4	Cathode Ray Oscilloscope	(0-20) MHz	1
5	Decade Resistance Box	10Ω-100KΩ	1
6	Connecting wires	5A	12

3.3 Theory:

During positive half-cycle of the input voltage, the diode D1 is in forward bias and conducts through the load resistor R1. Hence the current produces an output voltage across the load resistor R1, which has the same shape as the +ve half cycle of the input voltage.

During the negative half-cycle of the input voltage, the diode is reverse biased and there is no current through the circuit. i.e, the voltage across R1 is zero. The net result is that only the +ve half cycle of the input voltage appears across the load. The average value of the half wave rectified o/p voltage is the value measured on dc voltmeter.

For practical circuits, transformer coupling is usually provided for two reasons.

1. The voltage can be stepped-up or stepped-down, as needed.
2. The ac source is electrically isolated from the rectifier. Thus preventing shock hazards in the secondary circuit.

3.4 Procedure:

Half Wave Rectifier without filter

1. Connect the circuit as shown in figure (a).
2. Adjust the load resistance, RL to 500Ω, and note down the readings of input and output voltages through oscilloscope.
3. Note the readings of dc current, dc voltage and ac voltage.
4. Now, change the resistance the load resistance, RL to 1 KΩ and repeat the procedure as above. Also repeat for 10 KΩ.
5. Readings are tabulated as per the tabular column.

Half Wave Rectifier with filter

1. Connect the circuit as shown in figure (b) and repeat the procedure as for half wave rectifier without filter.

3.5 Circuit Diagrams:

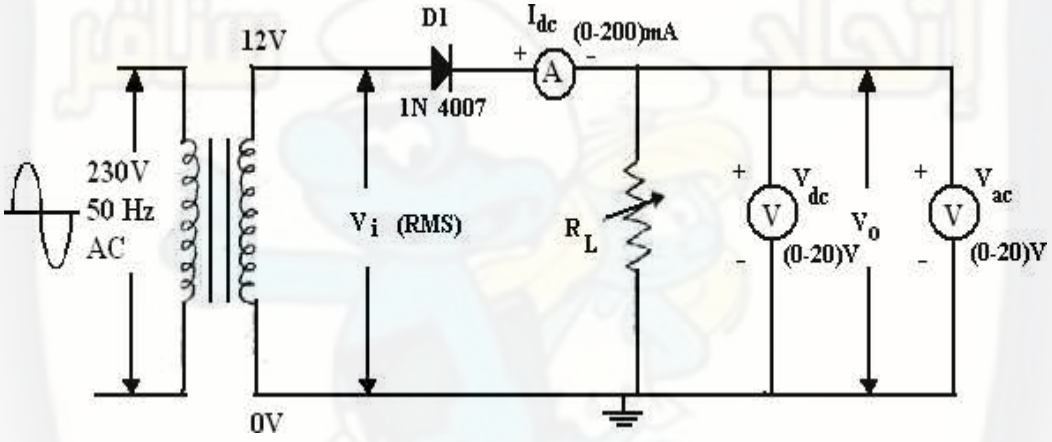


Figure (a): Half Wave Rectifier without Filter

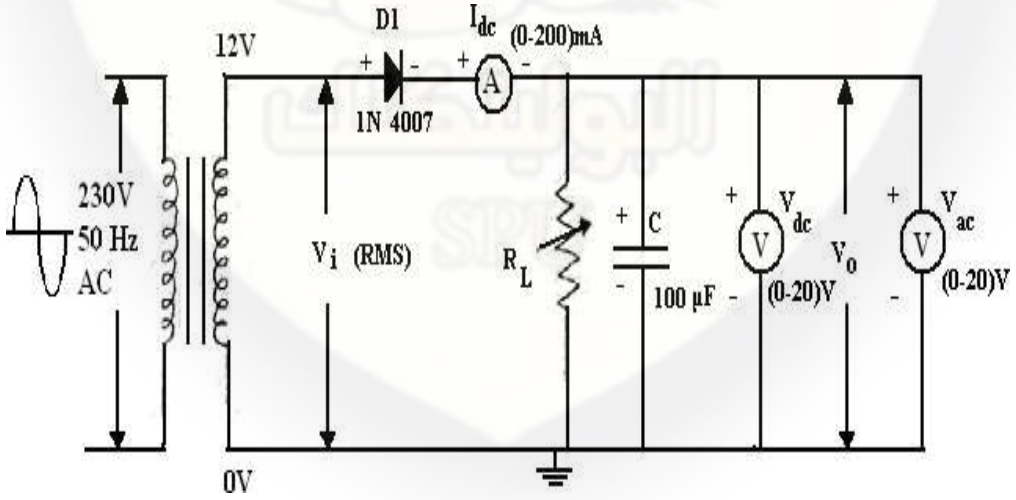
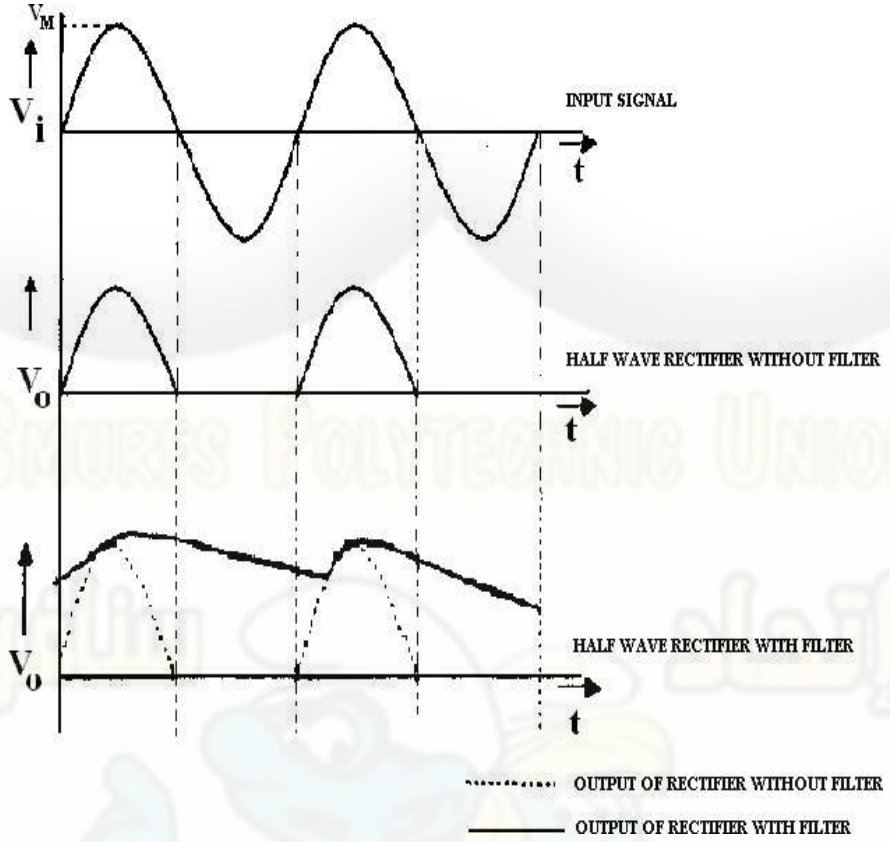


Figure (b): Half Wave Rectifier with Filter

3.6 Expected Graphs:



3.7 Tabular Column:

Half Wave Rectifier without Filter

#	Load Resistance (R_L)	Input Voltage Peak (V_m)	Output Voltage Peak (V_o)	Average dc current (I_{dc})	Average Dc voltage (V_{dc})	RMS Voltage (V_{ac})	Ripple Factor $\gamma = V_{ac}/V_{dc}$
1.	500Ω						
2.	1KΩ						
3.	10KΩ						

Half Wave Rectifier with Filter (C=10μF)

#	Load Resistance (R_L)	Input Voltage Peak (V_m)	Output Voltage Peak (V_o)	Average dc current (I_{dc})	Average Dc voltage (V_{dc})	RMS Voltage (V_{ac})	Ripple Factor $\gamma = V_{ac}/V_{dc}$
1.	1KΩ						
2.	2KΩ						
3.	10KΩ						

3.8 Precautions:

1. No loose contacts at the junctions.
2. Meters of correct ranges must be used for precision

3.9 Questions:

1. What is a rectifier?
2. How Diode acts as a rectifier?
3. What is the significance of PIV? What is the condition imposed on PIV?
4. What is meant by regulation?
5. What is meant by time constant?
6. Draw the o/p wave form without filter?
7. Draw the o/p wave form with filter?
8. What is meant by ripple factor? For a good filter whether ripple factor should be high or low?
9. What happens to the o/p wave form if we increase the capacitor value?
10. What happens if we increase the capacitor value?

3.10 Lab Assignment:

Plot the wave forms of Half wave rectifier with RL=5000 ohms, C = 680μF.

3.11 Result:

1. Input and Output waveforms of a half-wave with /without filter are observed and plotted.
2. For Half-wave rectifier without filter- γ, Ripple factor at 500Ω=
   1KΩ=
   10 KΩ=
3. For Half-wave rectifier with filter:- γ, Ripple factor at 1KΩ, 100μF =
   2KΩ, 100μF =
   10 KΩ, 100μF =

EXPERIMENT FOUR

FULLWAVE RECTIFIERS WITH/WITHOUT FILTER

4.1 Objectives:

– Examine the input and output waveforms of a full wave (center tapped) rectifier without and with filters. Calculate the ripple factor with load resistance of 500Ω, 1 KΩ and 10 KΩ respectively.

– Calculate ripple factor with a filter capacitor of 100μF and the load of 1KΩ, 2KΩ and 10KΩ respectively.

4.2 Components & Equipment required:

#	Device	Range/Rating	Quantity (in No.s)
1	Rectifier and Filter trainer Board Containing a) AC Supply. b) Silicon Diodes c) Capacitor	 (9-0-9) V 1N 4007 0.47μF	 1 2 1
2	a) DC Voltmeter b) AC Voltmeter	(0-20) V (0-20) V	1 1
3	DC Ammeter	(0-50) mA	1
4	Cathode Ray Oscilloscope	(0-20) MHz	1
5	Decade Resistance Box	10Ω-100KΩ	1
6	Electrolytic Capacitor	100μF	1
7	Connecting wires	5A	12

4.3 Theory:

The circuit of a center-tapped full wave rectifier uses two diodes D1&D2. During positive half cycle of secondary voltage (input voltage), the diode D1 is forward biased and D2is reverse biased.

The diode D1 conducts and current flows through load resistor R_L. During negative half cycle, diode D2 becomes forward biased and D1 reverse biased. Now, D2 conducts and current flows through the load resistor R_L in the same direction. There is a continuous current flow through the load resistor R_L, during both the half cycles and will get unidirectional current as show in the model graph. The difference between full wave and half wave rectification is that a full wave rectifier allows unidirectional (one way) current to the load during the entire 360 degrees of the input signal and half-wave rectifier allows this only during one half cycle (180 degree).

4.4 Procedure:

Full-wave Rectifier without filter

1. Connect the circuit as shown in the figure (a).
2. Adjust the load resistance RL to 500Ω and connect a capacitor of 100μF value in parallel with the load and note the readings of input and output voltages through Oscilloscope.
3. Note the readings of DC current, DC voltage and AC voltage.
4. Now change the load resistance RL to 2000Ω and repeat the procedure as the above.
5. Readings are tabulate as per the tabular column.

Full-wave Rectifier with filter

1. Connect the circuit as shown in the figure (b).
2. Adjust the load resistance RL to 1KΩ and connect a capacitor of 100μF values in parallel with the load and note the readings of input and output voltages through Oscilloscope.
3. Note the readings of DC current, DC voltage and AC voltage.
4. Now change the load resistance RL to 2KΩ and repeat the procedure as the above. Also repeat for 10K, 100μF values.
5. Readings are tabulate as per the tabular column.

4.5 Circuit Diagrams:

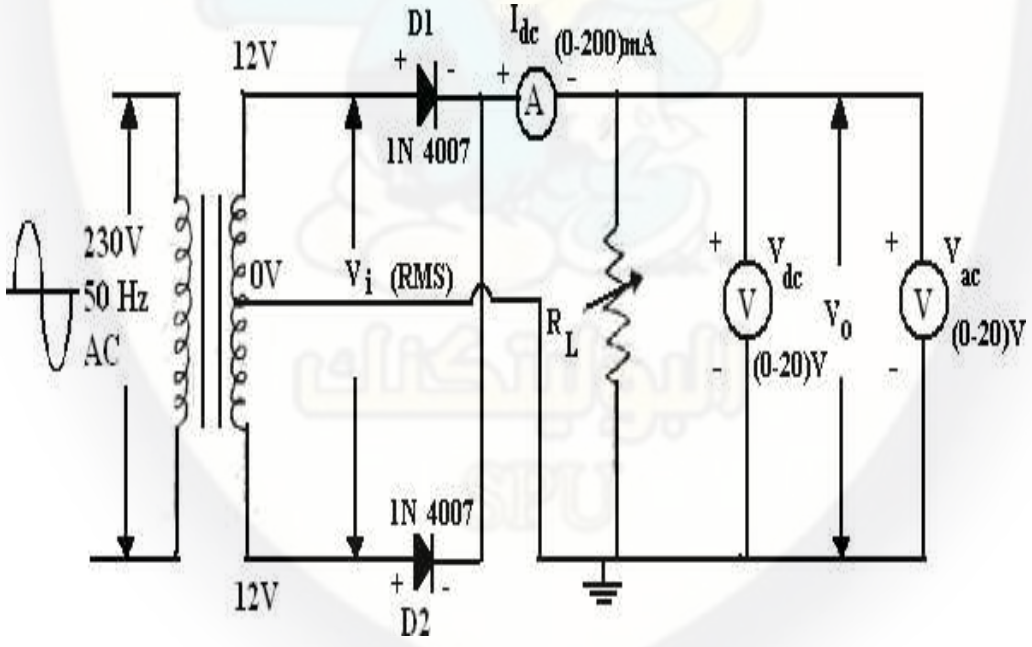


Figure (a): Full Wave Rectifier (Center-tap) Without Filter

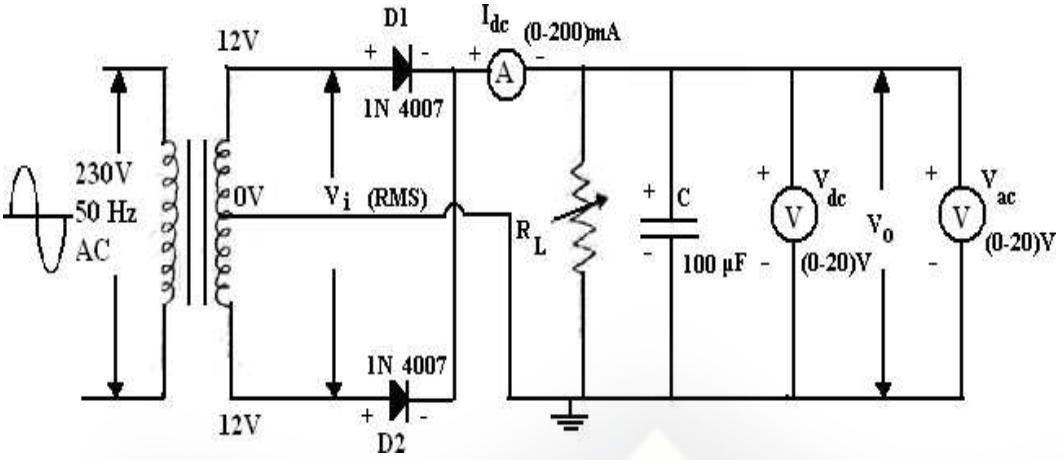
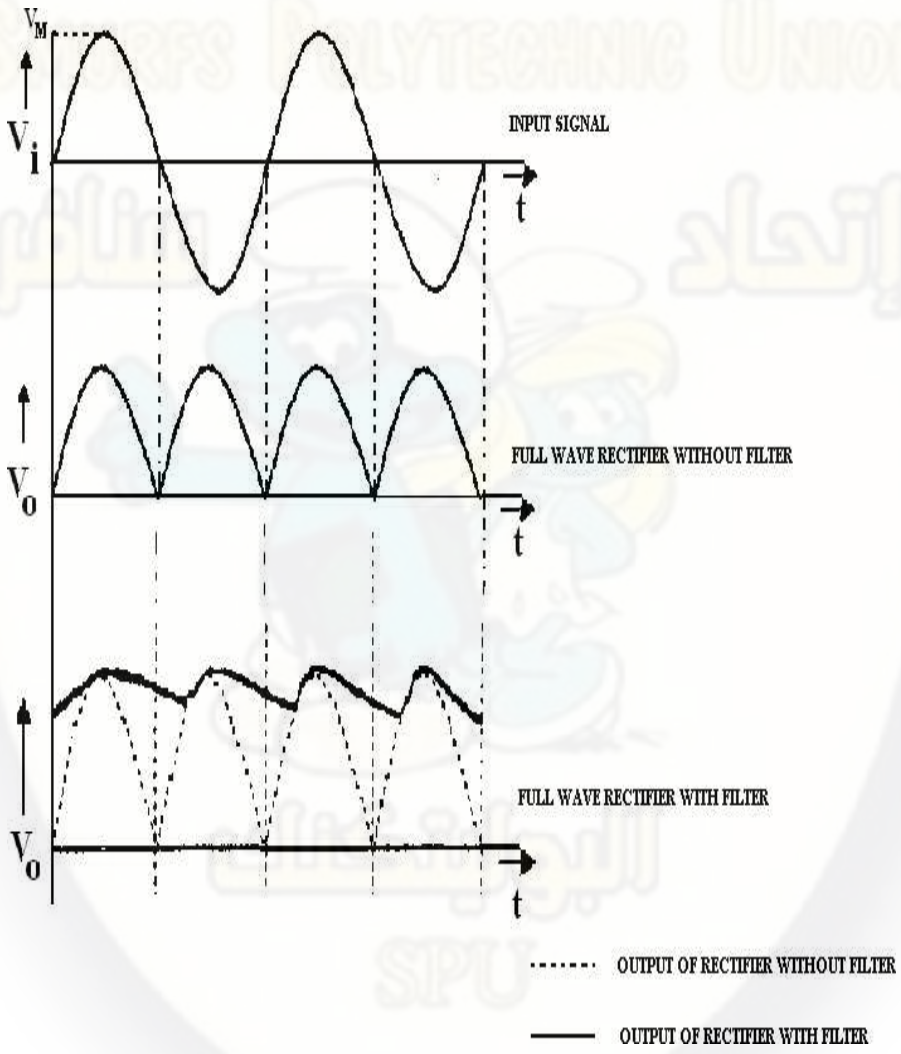


Figure (b): Full Wave Rectifier (center-tap) With Filter



4.6 Expected Graphs:

4.7 Tabular Column:

Full wave Rectifier (Center-tap) Without Filter

#	Load Resistance (R_L)	Input Voltage Peak (V_m)	Output Voltage Peak (V_o)	Average dc current (I_{dc})	Dc voltage (V_{dc})	RMS Voltage (V_{ac})	Ripple Factor $\gamma = V_{ac}/V_{dc}$
1.	500Ω						
2.	1KΩ						
3.	10KΩ						

Full wave Rectifier (Center-tap) With Filter C = --- µF

#	Load Resistance (R_L)	Input Voltage Peak (V_m)	Output Voltage Peak (V_o)	Average dc current (I_{dc})	Dc voltage (V_{dc})	RMS Voltage (V_{ac})	Ripple Factor $\gamma = V_{ac}/V_{dc}$
1.	500Ω						
2.	1KΩ						
3.	10KΩ						

4.8 Precautions:

1. No loose contacts at the junctions.
2. Meters of correct range must be used for precision.

4.9 Questions:

1. What is a full wave rectifier?
2. How Diode acts as a rectifier?
3. What is the significance of PIV requirement of Diode in full-wave rectifier?
4. Compare capacitor filter with an inductor filter?
5. What is the theoretical maximum value of ripple factor for a full wave rectifier?
6. Draw the o/p wave form without filter? Draw the O/P? What is wave form with filter?
7. What is meant by ripple factor? For a good filter whether ripple factor should be high or low? What happens to the ripple factor if we insert the filter?
8. What is meant by regulation? Why regulation is poor in the case of inductor filter?
9. What is meant by time constant?
10. What happens to the o/p wave form if we increase the capacitor value? What happens if we increase the capacitor value?

4.10 Lab Assignment:

Plot the wave forms of fulle wave rectifier with RL=500 ohms, C = 470μF.

4.11 Result:

1. Input and Output waveforms of a full-wave (center tapped) and bridge rectifier with /without filters are observed and plotted.
2. For Full-wave rectifier(center tapped) without filter-γ, Ripple factor at 500Ω, 100μF = 2KΩ, 100μF =
3. For full-wave rectifier (Center tapped) with filter-γ, Ripple factor at 500Ω, 100μF = 2KΩ, 100μF =

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EXPERIMENT FIVE

THE DIODE LIMITER

1.1 Objectives:

To demonstrate the operation of various diodes Limiter circuits.

1.2 Components & Equipment required:

- **Dual Trace Oscilloscope**
- **DMM** **2 IN 4001 Diodes**
- **Dual DC Power Supply.** **15KΩ Resistors**
- **Breadboard**
- **Function Generator** **Jumpers.**

1.3 Theory:

Clippers (Limiters) have the ability to limit "clip-off" a portion of the input signal without distorting the remaining part of the alternating waveform. The half-wave rectifier is an example of the series configuration of diode clipper. Depending on the direction of the diode, the positive or negative region of the input signal is “clipped” off. There are two general categories of clippers: *series* and *parallel*. The series configuration is defined as one where the diode is in series with the load, while the parallel variety has the diode in a branch parallel to the load. Figure (a) shows positive parallel limiter.

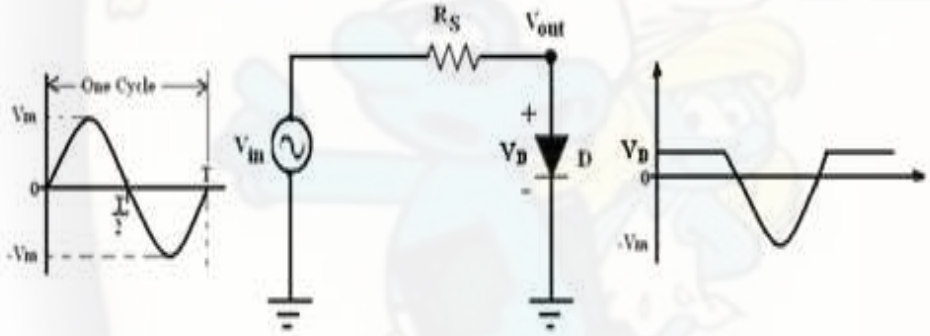


Figure (a): Parallel Limiter

The addition of a dc supply such as shown in Figure (b) can have a pronounced effect on the output of a clipper, and is known as biased limiter. V_{DC} in the voltage divider method is given by:

$$V_{DC} = \left(\frac{R_2}{R_1 + R} \right) * V_{CC}$$

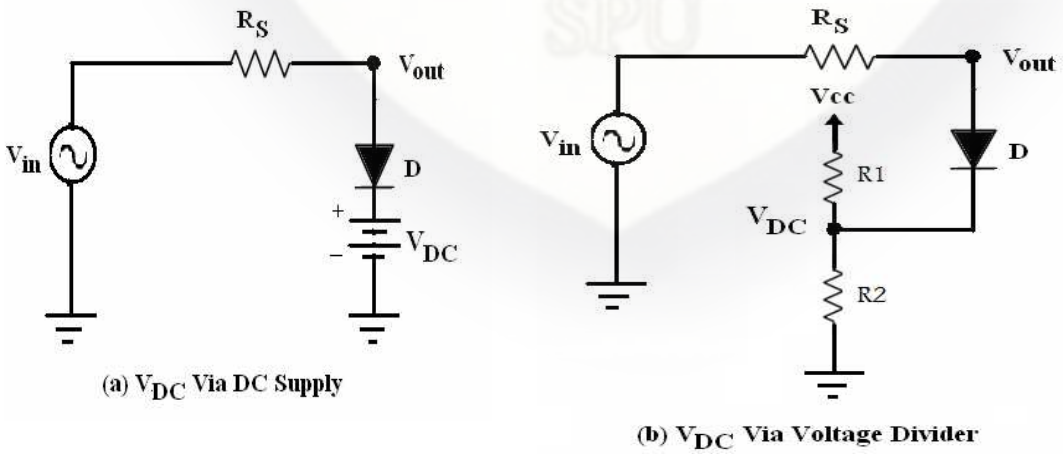


Figure (b): Biased Parallel limiters

1.4 PROCEDURE

1. Wire the circuit shown in Figure (c).
2. Connect CH1 at point A with respect to ground.
3. Adjust V_{in} to $6V_{p-p}$/200Hz sine wave.
4. Connect CH2 at point B with respect to ground to measure the limiting level.
5. On a graph draw both input and output signals.
6. Wire the circuit of figure (d). Gradually increase VDC and observe V_{out} . Explain your results and determine VDC value that will retain unclipped V_{out}.

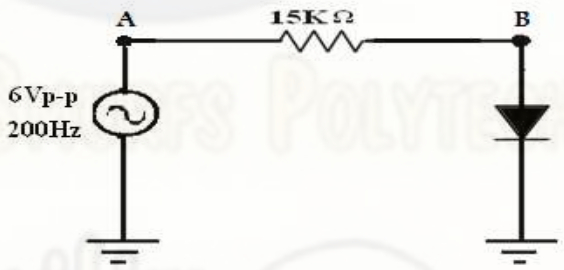


Figure (c): Basic Positive Limiter



Figure (d): Biased Limiter

7. Wire the circuit of figure (e) and repeat steps 4 and 5.

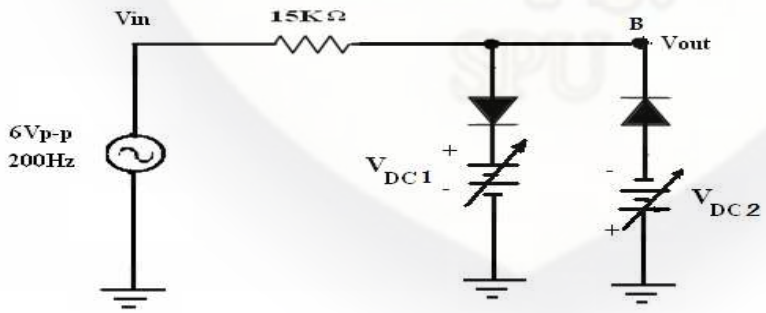


Figure (e): Double Biased Limiter

EXPERIMENT SIX

CLAMPERS AND VOLTAGE MULTIPLIER

1.1 Objectives:

To design Clampers and Voltage doubler for the given specification.

1.2 Components & Equipment required:

#	Device	Range/Rating	Quantity (in No.s)
1.	Diode (1N4007)	-	02
2.	Resistor & capacitors	As per design	-
3.	Multimeter	-	01

1.3 Procedure:

A. Clampers

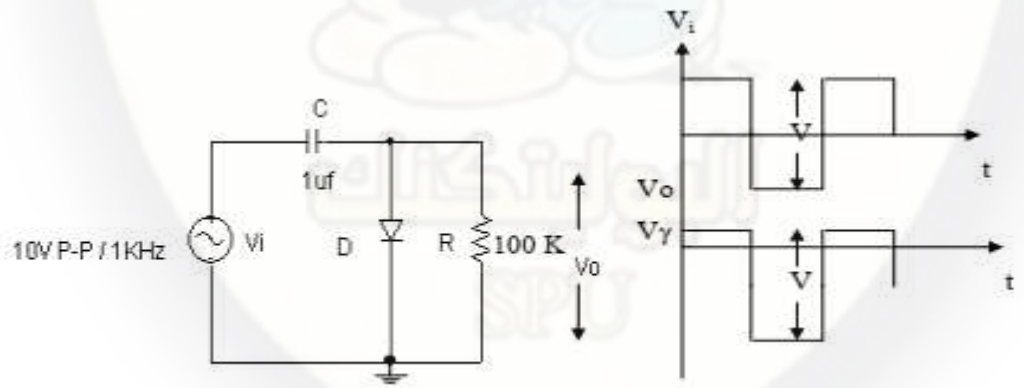
1. Connections are made as shown in the circuit diagram.
2. Apply a square wave / triangular wave / sine wave input of amplitude 10 V peak to peak and frequency of 1 kHz
3. Observe the input and output waveform.
4. Measure the clamping level and compare with the designed value

B. Voltage Multiplier

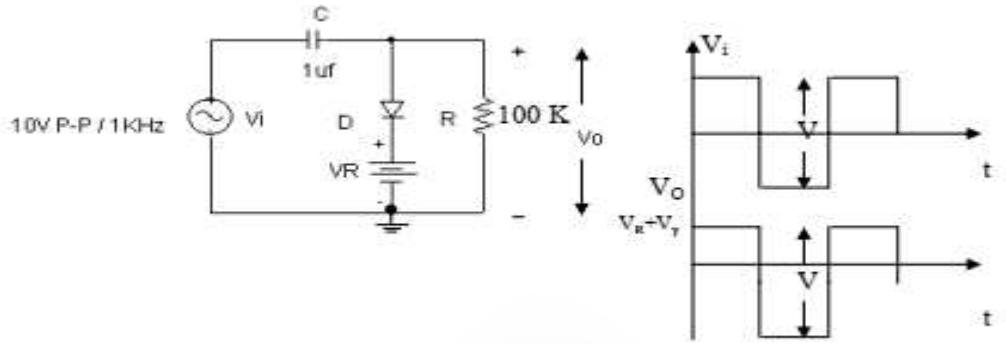
1. Connections are made as shown in the circuit diagram
2. Apply a square wave / triangular wave / sine wave input of amplitude 10 V peak to peak and frequency of 1 kHz
3. Observe the input and output waveform.
4. Measure the output voltage using multimeter in DC position.

***Negative Clampers*:**

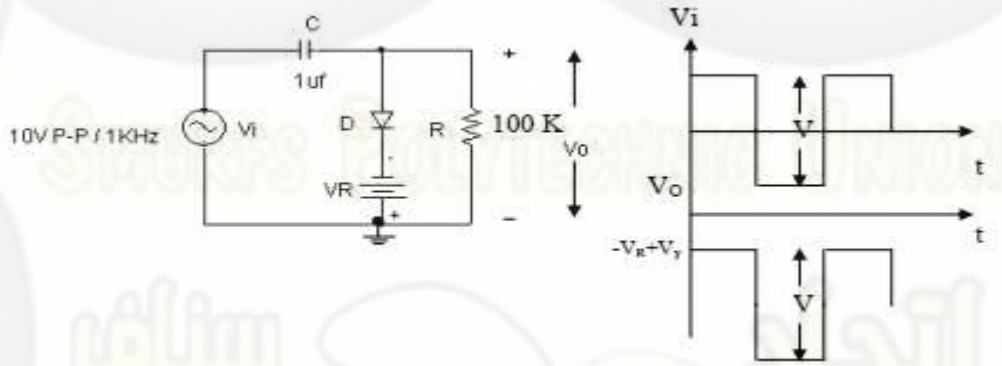
- Positive peak clamped to positive reference level (V_R+V_T)



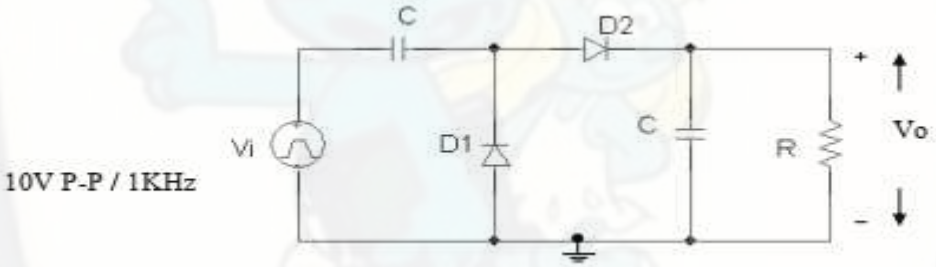
- Positive peak clamped to negative reference level (-V_R+V_T)



- Positive peak clamped to negative reference level (-V_R+V_T)



1.4 Circuit Diagram: Voltage Doubler



Design: Clampers

Example. 1: Design a clamping circuit to clamp the negative peak to +1V (Circuit 2)

$V_R - V_T = 1V$ Let the diode be silicon make then $V_T = 0.6V$ Then

$V_R = 1 + 0.6 = 1.6V$

For a clamper RC>>T, let RC = 100 T

Assume f = 1 kHz, hence T = 1 ms, choose C = 1µf Then R = 100 KΩ

Example. 2: Design a clamping circuit to clamp the positive peak to +2V (Circuit 3)

$V_R+V_T = 2V$ Let the diode be silicon make then $V_T = 0.6$ V $V_R = 2 - 0.6$

$V_R = 1.4$ V

For a clamper RC>>T, let RC = 100 T

Assume f = 1kHz, hence T = 1 ms, choose C = 1µf Then R = 100 kΩ

Design: Voltage Doubler

For Voltage doubler, RC>>T, let RC = 100 T

Assume f = 1 kHz, hence T = 1ms, choose C = 1 µf Then R = 100 K

1.5 Result:

#	Circuit	Designed Clamping Level	Observed Clamping Level
A. Positive Clampers			
1.	Negative peak clamped to $-V_T$ level	$-V_T$ =	
2.	Negative peak clamped to positive reference level (V_R-V_T)	V_R-V_T =	
3.	Negative peak clamped to negative reference level (-V_R-V_T)	-V_R-V_T =	
B. Negative Clampers			
1.	Positive peak clamped V_T level	V_T =	
2.	Positive peak clamped to positive reference level (V_R+V_T)	V_R+V_T =	
3.	Positive peak clamped to negative reference level (-V_R+V_T)	-V_R+V_T =	
C. Voltage Multiplier			
1.	Voltage Doubler	Vo = 2V_M - 2 V_T =	

EXPERIMENT SEVEN

ZENER CHARACTERISCTICS
& ZENER AS A VOLTAGE REGULATOR

1.1 Objectives

- To demonstrate the Zener diode characteristics in DC mode.
- To demonstrate the operation of Zener diode as a voltage regulator.

1.2 Components & Equipment required:

- **Function Generator** **6.2V Zener Diode.**
- **Dual Trace Oscilloscope**
- **DMM (Digital Multi Meter)** **(2* 220Ω), 100Ω , 10Ω resistors**
- **Breadboard**
- **Dual Dc power supply**

1.3 Theory:

Zener Diodes are basically the same as the standard PN junction diode but are specially designed to have a low pre-determined reverse breakdown voltage as shown in figure (a) that takes advantage of this high reverse voltage. Zener diodes are used as voltage regulators.

It had been seen that the Zener diode has a region in its reverse bias characteristics of almost a constant negative voltage regardless of the value of the current flowing through the diode and remains nearly constant even with large changes in current as long as the Zener diodes current remains between the breakdown current $I_{Z(min)}$ and the maximum current rating $I_{Z(max)}$. The ability of controlling the voltage can be used to great effect to regulate or stabilize a voltage source against supply or load variations, and the function of a regulator is to provide a constant output voltage to a load connected in parallel with it in spite of the ripples in the supply voltage or the variation in the load current and the Zener diode will continue to regulate the voltage until the diodes current falls below the minimum $I_{Z(min)}$ value in the reverse breakdown region.

Figure (a) is an analysis circuit for Zener diodes when are used as a voltage regulators. When E_S is less than Vz, Zener acts as an open circuit and V_O is given by:

$$V_O = (R_L/(R_L + R_S)E_S \quad (1)$$

And, $$I_L = V_L / R_L \quad (2)$$

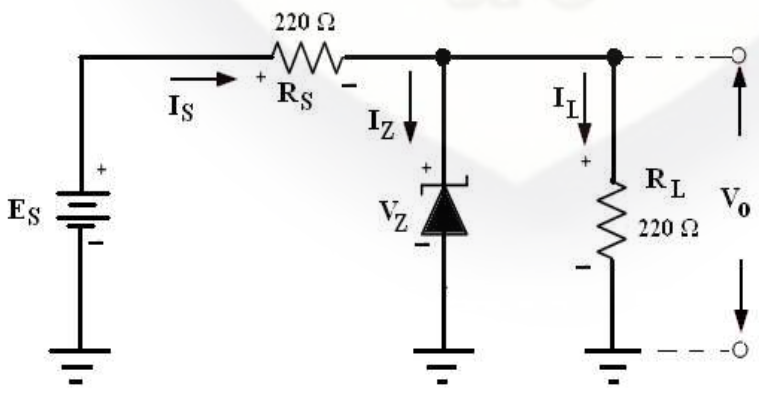


Figure (a): Analysis diagram for Zener diode as a regulator

When $E_S \geq V_Z$, then:

$$V_O = V_L = V_Z \quad (3)$$

And, $$I_S = I_Z + I_L \quad (4)$$

The suorce current (I_S) is given by:

$$I_S = (E_S - V_O)/R_S \quad (5)$$

And power dissipation(P_Z)is:

$$P_Z = I_Z V_Z \quad (6)$$

Regulation percentage (%V_O)

$$\%V_O = \left[(V_{NL} - V_{FL})/V_{FL}\right] * 100\% \quad (7)$$

Where, $\mathbf{V_{NL}}$ — Output voltage without Load, and $\mathbf{V_{FL}}$ — Output voltage with Load

1.4 Procedure:

PART 1: ZENER CHARACTERISTICS

1. Wire the network of figure (b).

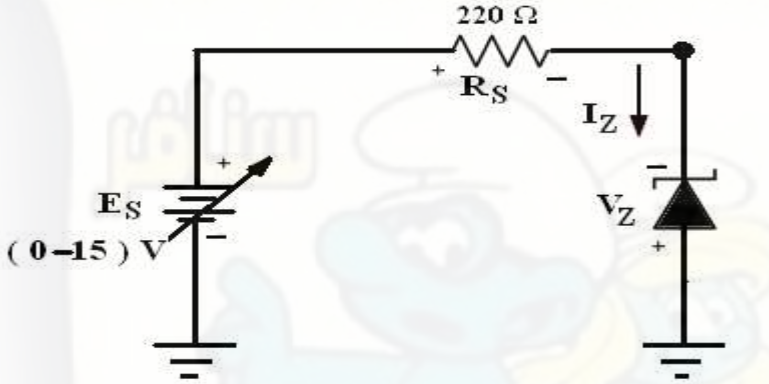


Figure (b): Zener Characteristics Circuit

2. Increase the dc supply voltage in small steps while simultaneously measuring the voltage across (**Vz**) and calculate the current (**Iz)** using (**Iz = V_{RS}/R_S**) through the Zener diode. In the vicinity of the Zener's knee voltage (approximately 6 V), make these steps approximately 0.05V ***Do not exceed a Zener current of 40 mA*.** Record your data in Table 7.1.
3. Plot your results for the corresponding Zener current and voltage values on the graph provided for this purpose.
4. At **I_Z** =20mA, determine **R_Z**,
5. Wire the circuit shown in figure (c). Calculate Iz, I_L and Vo. Record your results in table 4.2.

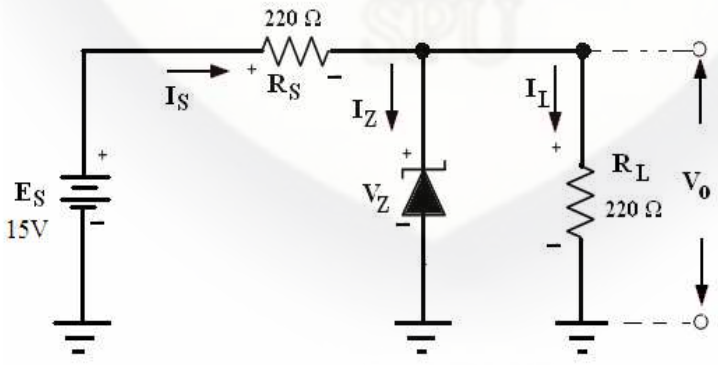


Figure (c): Analysis diagram for Zener diode as a regulator

6. Remove R_L, and calculate V_O.
7. Explain your results.

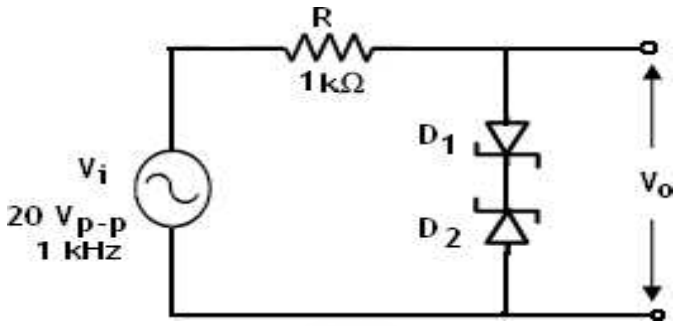
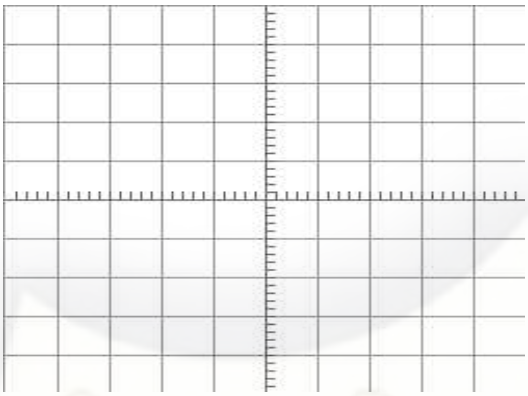


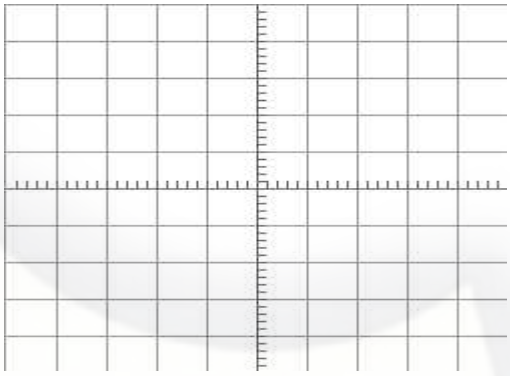
Figure (d): Zener Diodde Limiter

8. Connect the circuit as shown in Figure (d).
9. Apply sine wave of 20Vp-p, 1 kHz from function generator.
10. Observe the input, output waveforms and its transfer characteristics on CRO screen.

Volt/div = Time/div =



Graph 1: Output waveform for circuit in fig (d).



Graph 2: Transfer Characteristics (XY mode)

1.5 Tabular Column:

Table 7.1: Zener Characteristics

E_S (V)	V_{OUT} (V)	$I_Z = V_{RS}/R_S$ (mA)
5		
5.5		
6.00		
6.05		
6.10		
6.15		
6.20		
6.25		
6.30		
6.35		
6.40		
6.45		
6.55		
6.60		
7.00		
8.00		
9.0		
10		
11		
12		
13		
14		
15		

Table 7.2: data for Zener as Voltage Regulator

Parameter	Expected value	Measured value	% error
I_S			
I_Z			
I_L			
V_O (with Load)			
V_O (No Load)			

1- Replace Es by an Ac input voltage as shown in figure (e).
2- Set v_{in} to (10vp/100Hz). Draw the output and explain the results.

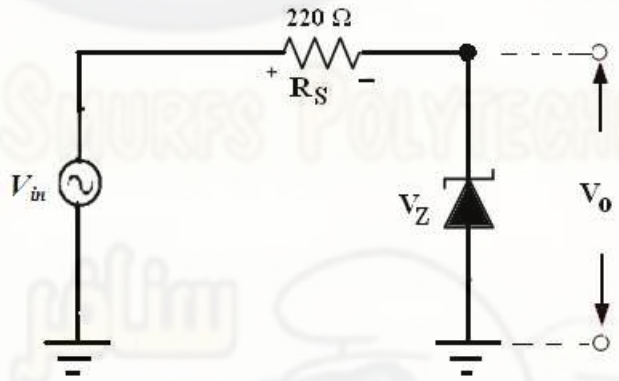


Figure (e): Zener With Ac input

EXPERIMENT EIGHT

BJT OPERATING REGIONS (Transistor as A Switch)
&
FIXED - BIASED CONFIGURATION

8.1 Objectives:

- To demonstrate the BJT operating regions the saturation, active and the cutoff regions.
- To demonstrate the fixed bias configuration as one of the transistor configuration methods.

8.2 Components & Equipment required:

- Dual DC power Supply
- DMM
- Breadboard
- Function Generator
- Dual Trace Oscilloscope.

1KΩ, 10KΩ, 100KΩ, 560KΩ
2N3904 npn silicon Transistor
Jumpers

8.3 Theory:

When operating the transistor, a dc biasing is necessary to establish the proper region of operation for ac amplification. The emitter layer is heavily doped, the base lightly doped, and the collector only lightly doped. The term bipolar reflects the fact that holes and electrons participate in the injection process into the oppositely polarized material. If only one carrier is employed (electron or hole), it is considered a unipolar device.

Transistor Biasing is the process of setting a transistor's DC operating voltage or current conditions to the correct level so that any AC input signal can be amplified correctly by the transistor. A transistor's steady state of operation depends a great deal on its base current, collector voltage, and collector current and therefore, if a transistor is to operate as a linear amplifier, it must be properly biased to have a suitable operating point. Establishing the correct operating point requires the proper selection of bias resistors and load resistors to provide the appropriate input current and collector voltage conditions. The correct biasing point for a bipolar transistor, either NPN or PNP, generally lies somewhere between the two extremes of operation with respect to it being either "fully-ON" or "fully-OFF" along its load line. This central operating point is called the "Quiescent Operating Point", or Q-point for short.

When a bipolar transistor is biased so that the Q-point is near the middle of its operating range, that is approximately halfway between cut-off and saturation, it is said to be operating as a Class-A amplifier. This mode of operation allows the output current to increase and decrease around the amplifiers Q-point without distortion as the input signal swings through a complete cycle. In other words, the output current flows for the full 360° of the input cycle.

The fixed-bias circuit of Figure (a) provides a relatively straightforward and simple introduction to transistor dc bias analysis. Even though the network employs an npn transistor, the equations and calculations apply equally well to a pnp transistor configuration merely by changing all current directions and voltage polarities.

8.4 Procedure:

Part A: The BJT Operating Regions

1. Construct the circuit of Figure 5.1. Adjust V_{BB} to 0V. This will set I_B = Zero,
2. Set V_{CC} to 5V, calculate and then measure I_C and V_{CE}. Record your results in table 8.1.
3. Set IB to 30μA, calculate and then measure IC and V_{CE}.
4. Set I_B to max by setting VBB to 6V, and still maintaining V_{CC} to 5V. Calculate and measure I_C and V_{CE}.

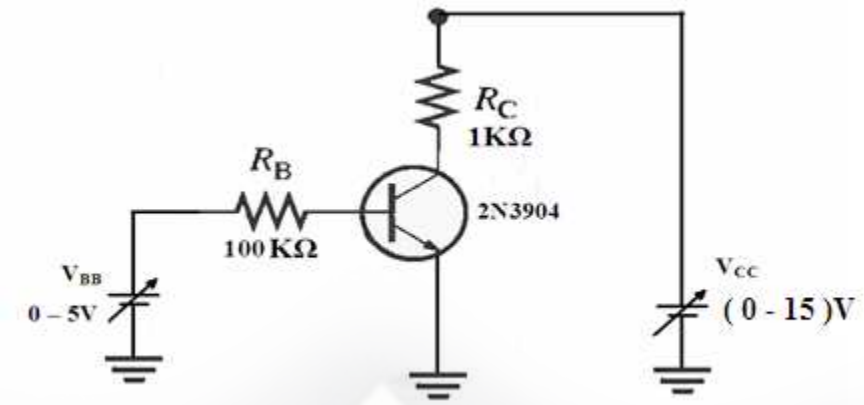


Figure (a): Collector Output Characteristics

5. Replace V_{BB} by (10Vp /100Hz) and V_{CC} to 5V as shown in figure (b).
6. Show and draw the output signal. Explain your results.

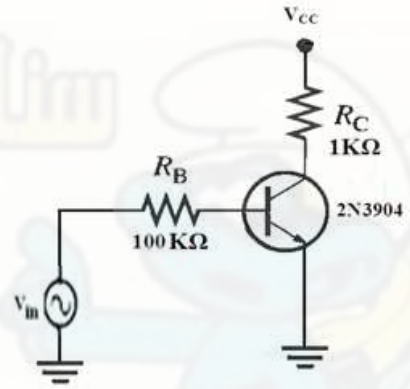


Figure (b): BJT Operating Regions

Part B: Fixed-Biased Configuration

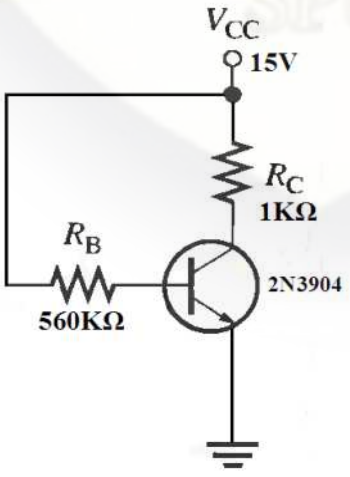
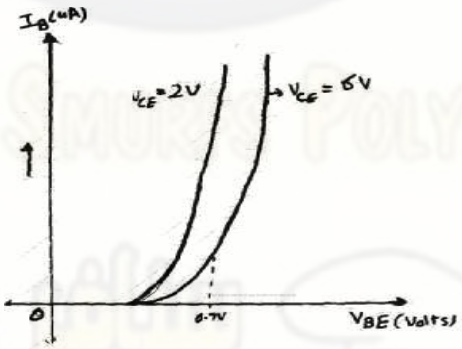


Figure (c): Fixed-Bias Configuration

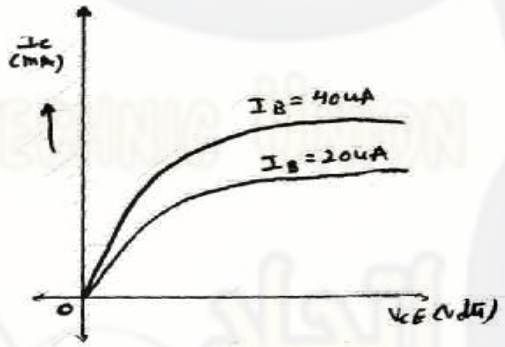
1. Construct circuit as of Figure 5.3 using 2N3904 transistor and set V_{CC} = 15 V.
2. Measure the voltages V_B and V_{RC}, and then record them in table 5.2.
3. Calculate the resulting base current, I_B and collector current, I_C. Using the values obtained, find β.
4. Using the values obtained in Step 3; calculate the values of V_B, V_C, V_E and V_{CE}.
5. Heat up the transistor in Figure 5.2, and then measure V_B, V_C, V_E and V_{CE}.
6. How do the measured values (Step 5) compare to the calculated values (Step 4)?

8.5 Expected Graphs:

Input Characteristics



Output Characteristics



8.6 Tabular Column:

Table 8.1: Collector Output Characteristics

I_B (μA)	Calculated		Measured	
	V_{CE}(V)	I_C(mA)	V_{CE}(V)	I_C(mA)
0				
30				
max				

Table 8.2: Self-Biased Configuration Results

Parameter	Transistor		Notes
	Expected Value	Measured Value	
I_B			
I_C			
β_{DC}			
V_{BE}	0.7V typical		
V_{CE}			

8.7 PRECAUTIONS

1. Keep the knobs of supply voltages V_{BE} & V_{CE} at minimum positions when switching ON or switching OFF the power supply.
2. No loose contacts at the junctions.
3. Do not overload the meters above its rated ranges.

EXPERIMENT NINE

TRANSISTOR CE CHARACTERISTICS

1.1 Objectives:

- Plot the input and output characteristics of a transistor connected in Common Emitter configuration.
- Calculate the input resistance R_i at I_B= 20 µA, output resistance R_o at V_{CE}=10V and current gain at V_{CE} =10V.

1.2 Components & Equipment required:

#	Device	Range/Rating	Quantity (in No.s)
1.	Transistor CE trainer Board Containing: a) DC Power Supply. b) PNP Transistor c) Carbon Film Resistor.	 (0-12) V BC 107 470Ω, 1/2 W 100KΩ,1/2W	 2 1 1
2.	a) DC Voltmeter b) DC Voltmeter	(0-1) V (0-20) V	1 1
3.	DC Ammeter	(0-50) mA (0-200) µA	1 1
4.	Connecting wires	5A	12

1.3 Theory:

A transistor is a three terminal device. The terminals are emitter, base, collector. In common emitter configuration, input voltage is applied between base and emitter terminals and output is taken across the collector and emitter terminals.

Therefore the emitter terminal is common to both input and output.

The input characteristics resemble that of a forward biased diode curve. This is expected since the Base-Emitter junction of the transistor is forward biased. As compared to CB arrangement I_B increases less rapidly with $V_{BE.}$ Therefore input resistance of CE circuit is higher than that of CB circuit.

The output characteristics are drawn between I_c and V_{CE} at constant $I_{B.}$ the collector current varies with V_{CE} unto few volts only. After this the collector current becomes almost constant, and independent of $V_{CE.}$ The value of V_{CE} up to which the collector current changes with V_{CE} is known as Knee voltage. The transistor always operated in the region above Knee voltage, I_C is always constant and is approximately equal to $I_{B.}$

The current amplification factor of CE configuration is given by

$$B = \Delta I_C / \Delta I_B$$

1.4 Procedure:

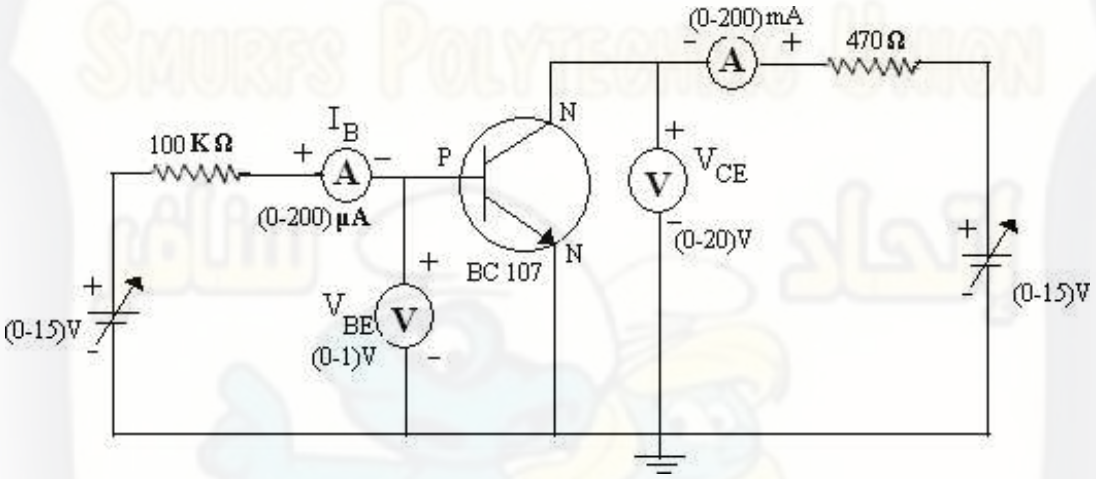
Input Characteristics:

1. Connect the transistor as shown in figure.
2. Keep the V_{CB} constant at 2V and 6V.
3. Vary the I_B in steps and note down the corresponding V_{EB} values as pertabular column.

Output Characteristics:

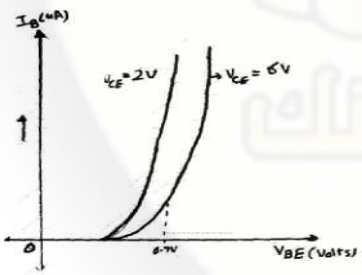
1. Keep the I_B constant at 20 μA and 40 μA.
2. Vary the V_{CE} in steps and note corresponding I_C values.
3. Readings are tabulated as shown in tabular column.

1.5 Circuit Diagram:

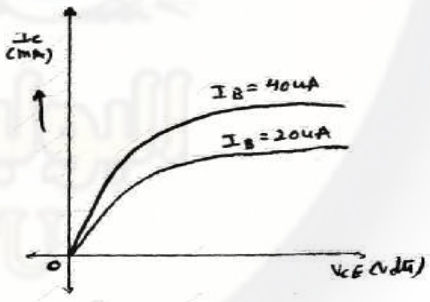


1.6 Expected Graphs:

Input Characteristics



Output Characteristics



1.7 Tabular Column:

Input Characteristics			
V_{CB} = 2V		V_{CB} = 6V	
V_{BE} (Volts)	I_B (μA)	V_{BE} (Volts)	I_B (μA)

Output Characteristics			
I_B = 20μA		I_B = 40μA	
V_{CE} (Volts)	I_C (mA)	V_{CE} (Volts)	I_C (mA)

1.8 Precautions:

1. Keep the knobs of supply voltages V_{BE} & V_{CE} at minimum positions when switching ON or switching OFF the power supply.
2. No loose contacts at the junctions.
3. Do not overload the meters above its rated ranges.

1.9 Calculations:

Input Resistance (I_B=20μA) = $\Delta V_{BE}/\Delta I_B$ = At V_{CE} = 2V
Input Resistance (I_B =20μA) = $\Delta V_{BE}/\Delta I_B$ = At V_{CE}= 6V
Output resistance (V_{CE}=10V) = $\Delta V_{CE}/\Delta I_C$ =
At I_B=20μA
Output resistance (V_{CE}=10V) = $\Delta V_{CE}/\Delta I_C$ =
At I_B=20μA
Current Amplification Factor "β" = $\Delta I_C/\Delta I_B$ =

H-PARAMETER CALCULATIONS

$h_{ie} = \Delta V_{be} / \Delta I_b$ =
$h_{oe} = \Delta I_c / \Delta V_{ce}$ =
$h_{fe} = \Delta I_c / \Delta I_b$ =
$h_{re} = \Delta V_{be} / \Delta V_{ce}$

1.10 Questions:

1. What is the range of ß for the transistor?
2. What are the input and output impedances of CE configuration?
3. Identify various regions in the output characteristics?
4. What is the relation between α *and* ß
5. Define current gain in CE configuration?
6. Why CE configuration is preferred for amplification?
7. What is the phase relation between input and output?
8. Draw diagram of CE configuration for PNP transistor?
9. What is the power gain of CE configuration?
10. What are the applications of CE configuration?

1.11 Lab Assignment:

Plot the I/O characteristics of CE configuration for Vcc = 10V, VBB = 4V, Rb= 200K ohms, Rc = 2K ohms, β = 200, Vbe = 0.7V.

1.12 Result:

1. Input and Output curves are plotted.
2. R_i, Input Resistance:
   a. V_{CE} = 2V and I_B =20 µA, R_i =
   b. V_{CE} = 6V and I_B =20 µA, R_i =
3. R_o, Output Resistance:
   a. V_{CE}= 10V and I_B = 20µA, R_o =
   b. V_{CE} = 10V and I_B = 40µA, R_o =
4. Current Amplification factor
   "β" = (at V_{CE} =10V)

EXPERIMENT ELEVEN

FET CHARACTERISTICS

11.1 Objectives:

- Find drain resistance at V_{DS} = 3V for V_{GS} = 0.5V, 0V, -1V.
- Find Trans conductance at V_{GS}= -1.0V for V_{DS}= 1V, 2V, 3V.

11.2 Components & Equipment required:

#	Device	Range / Rating	Quantity (in No.s)
1	FET Characteristics trainer board containing a) DC power supply b) DC power supply c) FET d) Carbon film resistor e) Carbon film resistor	 0-12V 0-5V BFW 11 1KΩ, 1/2W 470KΩ, 1/2W	 1 1 1 1 1
2	DC Voltmeter	0-20V	2
3	(a) DC Ammeter (b) DC Ammeter	0-200μA 0-20μA	1 1
4	Connecting wires		12

11.3 Theory:

A FET is a three terminal device, having the characteristics of high input impedance and less noise, the Gate to Source junction of the FET s always reverse biased. In response to small applied voltage from drain to source, the n-type bar acts as sample resistor, and the drain current increases linearly with $V_{DS.}$ With increase in I_D the ohmic voltage drop between the source and the channel region reverse biases the junction and the conducting position of the channel begins to remain constant. The V_{DS} at this instant is called “pinch of voltage”.

If the gate to source voltage (V_{GS}) is applied in the direction to provide additional reverse bias, the pinch off voltage ill is decreased.

In amplifier application, the FET is always used in the region beyond the pinch-off.

$F_{DS}=I_{DSS}(1-V_{GS}/V_P)^2$

11.4 Procedure:

Drain characteristics:

1. Connect the circuit as shown in figure (1)
2. Set the gate source voltage, V_{GS} in 0V position.
3. Increase drain source voltage, V_{DS} in steps and note corresponding I_D values as Shown in the tabular column.
4. Now change V_{GS} to +0.5Vand –1V and repeat the above procedure.

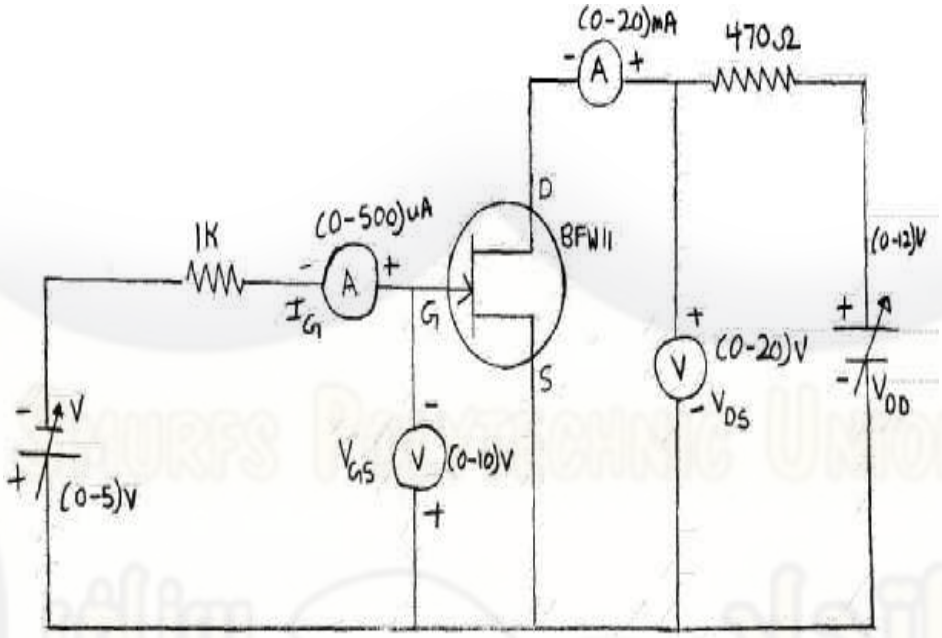
Transfer characteristics:

1. Connect the circuit as shown in figure (1).
2. Keep the drain source voltage, V_{DS} at 1V.
3. Vary the gate source voltage, V_{GS} in steps and note corresponding I_D values as shown in the

tabular column.

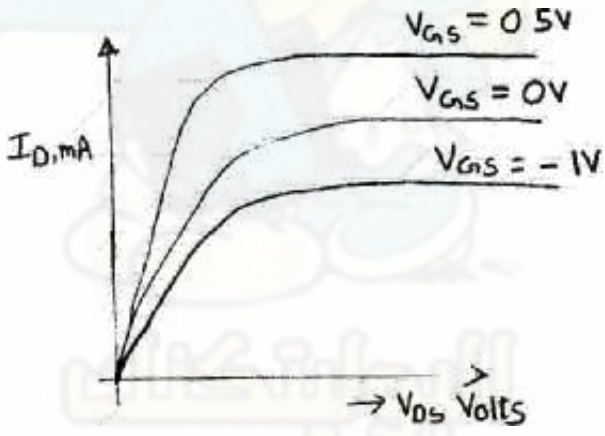
4. Now change V_{DS} to 2Vand 3V and repeat the above procedure.

11.5 Circuit Diagram:

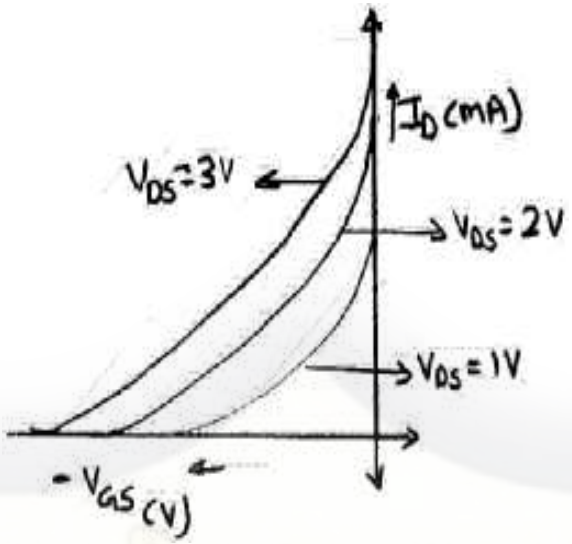


11.6 Expected Graphs:

- *Drain Characteristics*:



- *Transfer Characteristics*:



11.7 Tabular Column:

- *Drain characteristics:*

	V_{GS}=0V	V_{GS}= 0.5V	V_{GS}= 1.0V
V_{DS} (v)	**I_D (mA)**	**I_D (mA)**	**I_D (mA)**
0			
0.2			
0.4			
0.6			
0.8			
1.0			
2.0			
3.0			
4.0			
5.0			
6.0			
7.0			
8.0			
9.0			
10.0			

- *Transfer characteristics:*

	VDS=0V	VDS=2V	VDS= 3V
V_{GS} (v)	I_D (mA)	I_D (mA)	I_D (mA)
0			
-0.2			
-0.4			
-0.6			
-0.8			
-1.0			
-1.2			
-1.4			
-1.6			
-1.8			
-2.0			
-2.2			
-2.4			
-2.6			
-2.8			

11.8 Precautions:

1. Knobs of the supply must be kept at minimum position while switching on and off the supply.
2. Loose connections at junction must be avoided.
3. Device and meter of correct rating and range must be used.

11.9 Calculations:

- AC drain resistance at V_{DS}=3V of V_{GS}= 0.5V
  = $\Delta V_{DS}/ \Delta I_D$ ii. =
- AC drain resistance at V_{DS}=3V of V_{GS}= 0V
  = $\Delta V_{DS}/ \Delta I_D$ ii. =
- AC drain resistance at V_{DS}=3V of V_{GS}= -1V
  = $\Delta V_{DS}/ \Delta I_D$ ii. =
- Transconductance at V_{GS} = -1V of V_{DS}=1V
  = $\Delta I_D / \Delta V_{GS}$ ii. =
- Transconductance at V_{GS} = -1V of V_{DS}=2V
  = $\Delta I_D / \Delta V_{GS}$
- Transconductance at V_{GS} = -1V of V_{DS}=3V
  = $\Delta I_D / \Delta V_{GS}$

11.10 Questions:

1. What are the advantages of FET over transistor?
2. Is FET a current controlled device? Explain?
3. What is the operation of a N-channel JFET?
4. Can you compare JFET and a MOSFET?

11.11 Result:

1. At V_{DS}=3V
   - Drain Resistance for V_{GS}=0.5V is =
   - Drain Resistance for V_{GS}=0V is =
   - Drain Resistance for V_{GS}=0.5V is =
2. At V_{GS}=-1V
   - Transconductance for V_{DS} = 1V is =
   - Transconductance for V_{DS}= 2V is =
   - Transconductance for V_{DS}= 3V is =